Lecture V - Planning in Breweries

Applied Optimization with Julia

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Introduction

Case Study



- **Large brewery**
- Brews and sells beverages
- Production planning by hand
- Planner has a **lot of experience**
- **But** will retire soon

Challenges



- **Strong** competition
- Customer **demand is changing**

- Craft beer **gains popularity**
- **Variety** of drinks is increasing
- **Batch sizes are getting smaller**

Different costs



- Plant can fill **multiple types**
- Time depends on **type and batch**
- **Changing type** leads to setup costs for preparation and cleaning
- **Unsold beer bottles** can be **stored in a warehouse**
- This leads to **inventory costs**

Learning Objectives

After this lecture, you will be able to:

- **Formulate** the Capacitated Lot-Sizing Problem (CLSP)
- **Explain** the trade-off between setup and inventory holding costs
- **Recognize and construct** Big-M constraints
- **Discuss** the assumptions and limits of a finite planning horizon

One Bottling Line

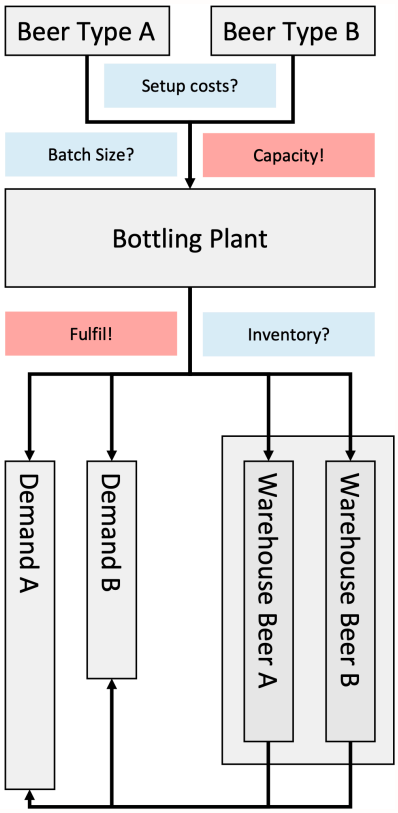
Question: Many beers share it. Where is the challenge?

Your Turn

Play: https://beyondsimulations.github.io/Applied-Optimization/games/bottling.html

Problem Structure

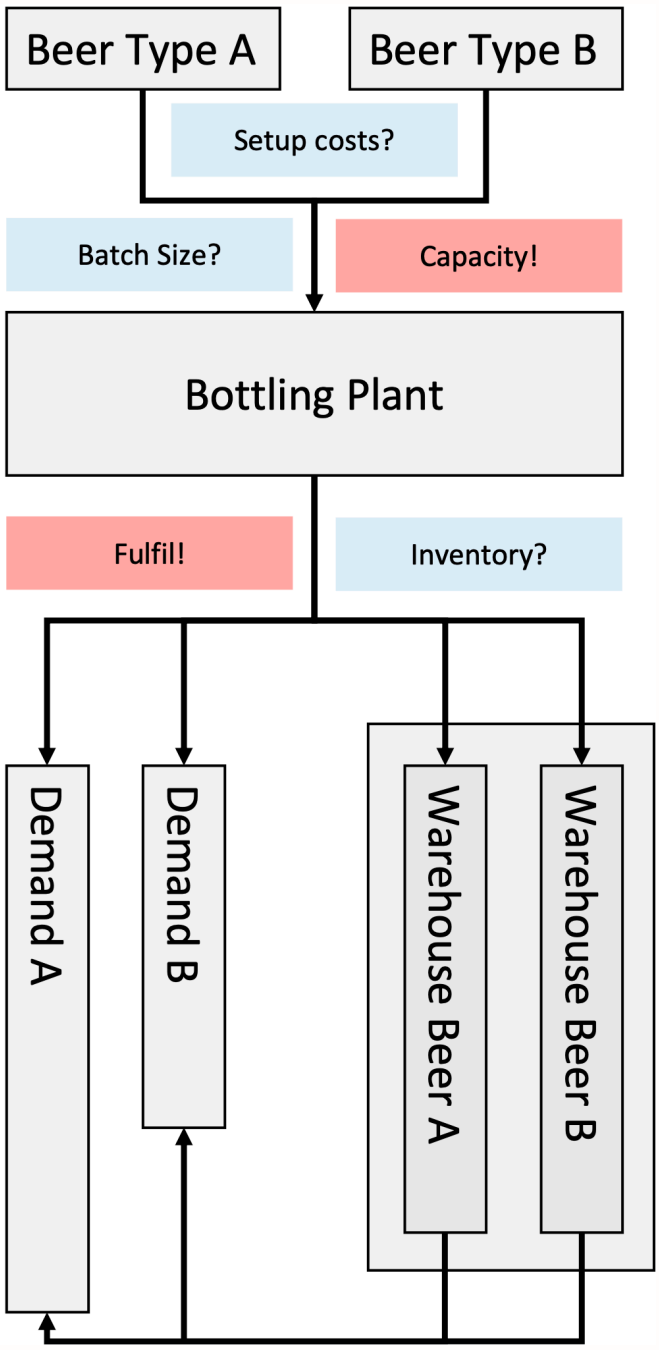
Objective



Question: What could be the objective?

Minimize the combined setup and inventory holding cost while satisfying the demand and adhering to the production capacity.

Trade-Off



Question: What is the trade-off?

Larger batches require **less setup cost per bottle**, but increase the storage cost.

Available Sets

Question: What are sets again?

Sets are **collections of objects**.

Question: What could be the sets here?

- $\mathcal{I}$ - Set of beer types indexed by $i \in \{1, 2, ..., |\mathcal{I}|\}$
- $\mathcal{T}$ - Set of time periods indexed by $t \in \{1, 2, ..., |\mathcal{T}|\}$

Available Parameters

Question: What are possible parameters?

- a_t - Available time on the bottling plant in period $t \in \mathcal{T}$
- b_i - Time used for bottling one unit of beer type $i \in \mathcal{I}$
- g_i - Setup time for beer type $i \in \mathcal{I}$
- f_i - Setup cost of beer type $i \in \mathcal{I}$
- c_i - Inventory holding cost for one unit of beer type $i \in \mathcal{I}$
- $d_{i,t}$ - Demand of beer type $i \in \mathcal{I}$ in period $t \in \mathcal{T}$

Decision Variables?

> **i We have the following sets:**
> - Beer types indexed by $i \in \{1, 2, ..., |\mathcal{I}|\}$
> - Time periods of the planning horizon indexed by $t \in \{1, 2, ..., |\mathcal{T}|\}$

> **! Our objective is to:**
> Minimize the combined setup and inventory holding cost while satisfying the demand and adhering to the production capacity.

Question: What could be our decision variable/s?

Decision Variables

- $W_{i,t}$ - Inventory of type $i \in \mathcal{I}$ at the end of $t \in \mathcal{T}$
- $Y_{i,t}$ - 1, if type $i \in \mathcal{I}$ is bottled in $t \in \mathcal{T}$, 0 otherwise
- $X_{i,t}$ - Batch size of type $i \in \mathcal{I}$ in $t \in \mathcal{T}$

Model Formulation

Objective Function?

> **! Our objective is to:**
> Minimize the combined setup and inventory holding cost while satisfying the demand and adhering to the production capacity.

Question: What could be our objective function?

We need the following variables:

- $W_{i,t}$ - Inventory of type $i \in \mathcal{I}$ at the end of $t \in \mathcal{T}$
- $Y_{i,t}$ - 1, if type $i \in \mathcal{I}$ is bottled in $t \in \mathcal{T}$, 0 otherwise

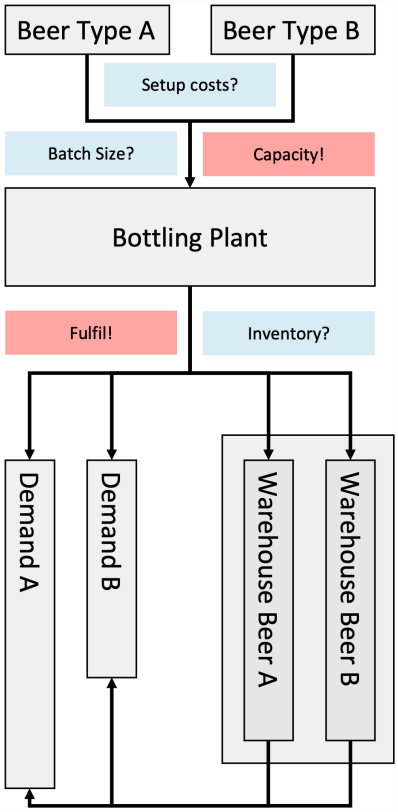
Objective Function

We need the following parameters:

- f_i - Setup cost of beer type $i \in \mathcal{I}$
- c_i - Inventory holding cost for one unit of beer type $i \in \mathcal{I}$

$$\text{Minimize} \quad \sum_{i \in \mathcal{I}} \sum_{t \in \mathcal{T}} (c_i \times W_{i,t} + f_i \times Y_{i,t})$$

Constraints



Question: What constraints?

- Transfer **unused** inventory
- **Fulfill** the customer demand
- **Set up** beer types
- Calculate **batch size** per set-up
- Compute **remaining** inventory
- **Limit** the bottling plant

Demand/Inventory Constraints?

! **The goal of these constraints is to:**
Consider the current inventory and batch sizes and compute the remaining inventory.

i **We need the following variables and parameters:**
- $W_{i,t}$ - Inventory of beer type $i \in \mathcal{I}$ at the end of period $t \in \mathcal{T}$
- $X_{i,t}$ - Batch size of beer type $i \in \mathcal{I}$ in $t \in \mathcal{T}$
- $d_{i,t}$ - Demand of beer type $i \in \mathcal{I}$ in period $t \in \mathcal{T}$

Question: What could the constraint look like?

Demand/Inventory Constraints

$$W_{i,t-1} + X_{i,t} - W_{i,t} = d_{i,t} \quad \forall i \in \mathcal{I}, t \in \mathcal{T}$$

with $W_{i,0} = 0$, as the warehouse starts empty.

i **Remember, these are the variables and parameters:**
- $W_{i,t}$ - Inventory of beer type $i \in \mathcal{I}$ at the end of period $t \in \mathcal{T}$
- $X_{i,t}$ - Batch size of beer type $i \in \mathcal{I}$ in $t \in \mathcal{T}$
- $d_{i,t}$ - Demand of beer type $i \in \mathcal{I}$ in period $t \in \mathcal{T}$

Why Fix the Starting Inventory?

Question: Why do we have to fix $W_{i,0} = 0$?

Otherwise, the solver could invent free starting inventory and would never need to bottle anything for the first weeks!

Setup Constraints?

! **The goal of these constraints is to:**
Set up beer types in all periods where the batch size is > 0.

i **We need the following variables and parameters:**
- $Y_{i,t}$ - 1, if beer type $i \in \mathcal{I}$ is bottled in period $t \in \mathcal{T}$, 0 otherwise
- $X_{i,t}$ - Batch size of beer type $i \in \mathcal{I}$ in $t \in \mathcal{T}$
- $d_{i,t}$ - Demand of beer type $i \in \mathcal{I}$ in period $t \in \mathcal{T}$

Question: What could the second constraint be?

Setup Constraints

$$X_{i,t} \leq Y_{i,t} \times \sum_{\tau \in \mathcal{T}} d_{i,\tau} \quad \forall i \in \mathcal{I}, t \in \mathcal{T}$$

Question: Do you know this type of constraint?

This type of constraint is called a **"Big-M"** constraint!

- **M** (here $\sum_{\tau \in \mathcal{T}} d_{i,\tau}$) is a large number
- It is coupled with a binary variable (here $Y_{i,t}$)
- **Like an if-then constraint**

Capacity Constraints?

> ! **The goal of these constraints is to:**
> Limit the capacity of the bottling plant per period.

> i **We need the following variables and parameters:**
> - $Y_{i,t}$ - 1, if beer type $i \in \mathcal{I}$ is bottled in period $t \in \mathcal{T}$, 0 otherwise
> - $X_{i,t}$ - Batch size of beer type $i \in \mathcal{I}$ in $t \in \mathcal{T}$
> - a_t - Available time on the bottling plant in period $t \in \mathcal{T}$
> - b_i - Time used for bottling one unit of beer type $i \in \mathcal{I}$
> - g_i - Setup time for beer type $i \in \mathcal{I}$

Capacity Constraints

Question: What could the third constraint be?

It uses more parameters than the previous constraints, but it is easier to understand.

$$\sum_{i \in \mathcal{I}} (b_i \times X_{i,t} + g_i \times Y_{i,t}) \leq a_t \quad \forall t \in \mathcal{T}$$

And that's basically it!

CLSP: Objective Function

The complete model is known as the **Capacitated Lot-Sizing Problem (CLSP)**.

$$\text{Minimize} \quad \sum_{i \in \mathcal{I}} \sum_{t \in \mathcal{T}} (c_i \times W_{i,t} + f_i \times Y_{i,t})$$

> ! **The goal of the objective function is to:**
> Minimize the combined setup and inventory holding cost while satisfying the demand and adhering to the production capacity.

CLSP: Constraints

$$W_{i,t-1} + X_{i,t} - W_{i,t} = d_{i,t} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (W_{i,0} = 0)$$

$$X_{i,t} \leq Y_{i,t} \times \sum_{\tau \in \mathcal{T}} d_{i,\tau} \quad \forall i \in \mathcal{I}, t \in \mathcal{T}$$

$$\sum_{i \in \mathcal{I}} (b_i \times X_{i,t} + g_i \times Y_{i,t}) \leq a_t \quad \forall t \in \mathcal{T}$$

> ! **Our constraints ensure:**
> Demand is met, inventory transferred, setup taken care of, and capacity respected.

CLSP: Variable Domains

$$Y_{i,t} \in \{0,1\} \quad \forall i \in \mathcal{I}, t \in \mathcal{T}$$

$$W_{i,t}, X_{i,t} \geq 0 \quad \forall i \in \mathcal{I}, t \in \mathcal{T}$$

> ! **The variable domains make sure that:**
> The binary setup variable is either 0 or 1 and that the inventory and batch size are non-negative.

Question: Why is a continuous batch size acceptable here?

Rounding a batch of thousands barely changes the cost.

Julia Implementation

Variables in Julia

Question: How could we formulate the variables in Julia?

```
using JuMP, HiGHS
beer_types = ["IPA", "Lager", "Stout"] # Beer types
periods = 1:3 # Time periods (weeks)
clsp_model = Model(HiGHS.Optimizer)
set_silent(clsp_model) # Hide the solver output

@variable(clsp_model, Y[i in beer_types, t in periods], Bin)
@variable(clsp_model, X[i in beer_types, t in periods] >= 0)
@variable(clsp_model, W[i in beer_types, t in 0:last(periods)] >= 0)
```

```
2-dimensional DenseAxisArray{JuMP.VariableRef,2,...} with index sets:
    Dimension 1, ["IPA", "Lager", "Stout"]
    Dimension 2, 0:3
And data, a 3×4 Matrix{JuMP.VariableRef}:
 W[IPA,0]    W[IPA,1]    W[IPA,2]    W[IPA,3]
 W[Lager,0]  W[Lager,1]  W[Lager,2]  W[Lager,3]
 W[Stout,0]  W[Stout,1]  W[Stout,2]  W[Stout,3]
```

Note, how we declare the inventory W also for period 0!

Objective Function in Julia

$$\text{Minimize} \quad \sum_{i \in \mathcal{I}} \sum_{t \in \mathcal{T}} (c_i \times W_{i,t} + f_i \times Y_{i,t})$$

```
c = Dict("IPA" => 0.1, "Lager" => 0.1, "Stout" => 0.1) # Holding costs
f = Dict("IPA" => 5000, "Lager" => 4000, "Stout" => 6000) # Setup costs

@objective(clsp_model, Min,
    sum(c[i] * W[i,t] + f[i] * Y[i,t]
        for i in beer_types, t in periods)
)
```

Data for our Small Brewery

We need the demand $d_{i,t}$, capacity a_t, and times b_i, g_i:

```
d = Dict( # Demand per beer type and week
    "IPA"   => [300, 800, 500],
    "Lager" => [600, 500, 800],
    "Stout" => [200, 300, 250],
)
a = [1800, 1800, 1800] # Capacity per week in minutes
b = Dict("IPA" => 0.9, "Lager" => 0.6, "Stout" => 1.2) # Min. per bottle
g = Dict("IPA" => 60, "Lager" => 40, "Stout" => 80) # Setup in minutes
```

Constraints in Julia

The constraints map almost one-to-one from the math:

```
@constraint(clsp_model, [i in beer_types],
    W[i,0] == 0) # The warehouse starts empty
@constraint(clsp_model, [i in beer_types, t in periods],
    W[i,t-1] + X[i,t] - W[i,t] == d[i][t])

@constraint(clsp_model, [i in beer_types, t in periods],
    X[i,t] <= sum(d[i]) * Y[i,t]) # Big-M setup constraint
@constraint(clsp_model, [t in periods],
    sum(b[i] * X[i,t] + g[i] * Y[i,t] for i in beer_types) <= a[t])
```

Solving the Model

```
optimize!(clsp_model)
println("Status: ", termination_status(clsp_model))
println("Total cost: ", objective_value(clsp_model))
```

```
Status: OPTIMAL
Total cost: 28130.0
```

Of the 28,130 total cost, 28,000 are setup costs and only 130 are inventory holding costs. But what does the plan look like?

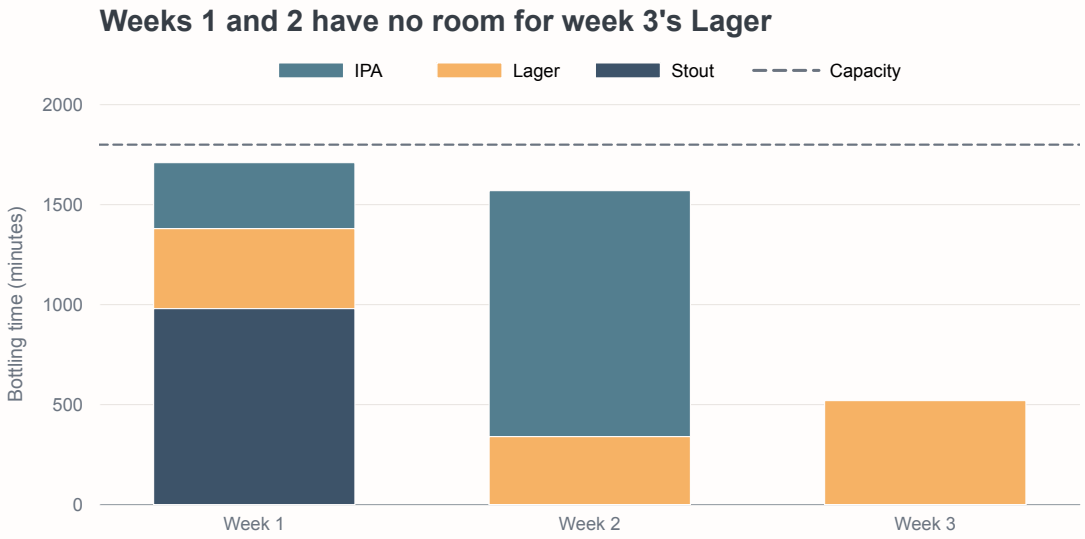
The Production Plan

```
for i in beer_types
    println(rpad(i, 8),
        "X = ", rpad(string([round(Int, value(X[i,t])) for t in periods]), 18),
        "W = ", [round(Int, value(W[i,t])) for t in periods])
end
```

```
IPA      X = [300, 1300, 0]    W = [0, 500, 0]
Lager    X = [600, 500, 800]   W = [0, 0, 0]
Stout    X = [750, 0, 0]       W = [550, 250, 0]
```

- **Stout** is bottled **only once**: week 1 covers all three weeks, saving two setups at 6,000 each
- **Lager** is bottled every week, as the plant capacity is too tight to batch it
- **IPA** batches the demand of week 3 into week 2

Where the Minutes Go



Model Characteristics

Recap on some Basics

There exist several types of optimization problems:

- **Linear (LP):** Linear constraints and objective function

- **Mixed-integer (MIP):** Linear constraints and objective function, but discrete variable domains
- **Quadratic (QP):** Quadratic constraints and/or objective
- **Non-linear (NLP):** Non-linear constraints and/or objective
- **And more!**

Recap on Solution Algorithms

- **Simplex algorithm** to solve LPs
- **Branch & Bound** to solve MIPs
- **Outer-Approximation** for mixed-integer NLPs
- **Matheuristics** (e.g., Fix-and-Optimize, ...)
- **Metaheuristics** (e.g., tabu search, genetic algorithms, ...)
- **Decomposition** methods (Lagrange, Benders, ...)
- **Heuristics** (greedy, construction method, n-opt, ...)
- **Graph theoretical methods** (network flow, shortest path)

Model Characteristics

Questions: On model characteristics

- Is the model formulation linear/ non-linear?
- What kind of variable domains do we have?
- What kind of solver could we use?
- Can the Big-M constraint be tightened?

Tightening the Big-M

Question: How small can we make M without cutting off solutions?

- Production in period t only serves demand from t onward: $M_{i,t} = \sum_{\tau \in \mathcal{T} | \tau \geq t} d_{i,\tau}$
- Capacity limits a batch as well: $M_{i,t} = (a_t - g_i)/b_i$

The **smaller** of the two is the tighter bound, and tighter formulations are usually solved faster!

Model Assumptions

Questions: On model assumptions

- What assumptions have we made?
- What is the problem with the planning horizon?
- Any idea how to solve it?

> **End-of-horizon effect**
> The model plans no inventory for demand after week $|\mathcal{T}|$, as it does not know about it. In practice, this is solved by re-planning on a **rolling horizon**.

Impact

Can this be

applied?

Scale as a Problem

Solving the problem with commercial solvers is not tractable, as it takes too long.

Scale of the Case Study

- **220** finished products
- **100** semi-finished products
- **13** production resources
- **8** storage resources
- **3** main production levels
- **52** weeks planning horizon

Our CLSP is the **single-level core**: the real case is multi-level, with semi-finished products and storage.

Any idea what

could be done?

Heuristics and Optimization

- Multi-level Capacitated Lot-Sizing Problem
- Heuristic fix and optimize approach[1]
- Operating cost reduction by 5% and planning effort by 40%

> **And that's it for today's lecture!**
> We now have covered the basics of the CLSP and are ready to start solving some tasks in the upcoming tutorial.

Questions?

[1] M. Mickein, M. Koch, and K. Haase [1]

Literature

For more interesting literature to learn more about Julia, take a look at the literature list of this course.

[1] M. Mickein, M. Koch, and K. Haase, “A Decision Support System for Brewery Production Planning at Feldschlösschen,” *INFORMS Journal on Applied Analytics*, vol. 52, no. 2, pp. 158–172, 2022.